

Numerical simulation of deuterium retention in tungsten under ELM-like conditions

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ABSTRACT

Retention of hydrogen isotopes (HI) in plasma-facing components (PFCs) is a crucial process influencing the operation of a fusion device. The dynamics of HI retention is mainly determined by irradiation conditions of PFCs. These conditions can change due to the onset of fast transient events, such as edge-localised modes (ELMs). The development of ELMs results in repetitive short-term plasma bursts on PFCs, affecting the exposure regime. In this work, the effect of ELM-like loads on the fuel retention is numerically analysed using a one-dimensional diffusion model, implemented in the FESTIM code. As a representative simulation case, the deuterium (D) retention in tungsten (W) is considered with the geometry and inter-ELM exposure conditions relevant for large fusion devices. Temporal dependencies of heat and particle fluxes during intra-ELM stages are accounted for by applying the free-streaming model for an ELM filament transport. The simulations were conducted for three inter-ELM exposure regimes with various properties of transient loads, D trapping sites, and D recombination rates. Compared to the ELM-free irradiation, the onset of transients is shown to mainly reduce the D retention rate because of significant material heating induced by the arrival of energetic ELM particles. This relative difference can decrease if the material is characterised by strong trapping sites or limited desorption from a surface. The findings also demonstrate that additional heating during transients provides conditions for a faster D migration into the bulk. In addition, we discuss the possibility of using ELM-average loads to obtain quick assessments of the D content. The approach is shown to be applicable for the case of small ELMs and is used to derive the analytical expressions for the D distribution in W within the steady-state approximation.

1. Introduction

In future fusion reactors, plasma-facing components (PFCs) may have to withstand intense ion and neutron irradiation. For normal operational regimes, the highest ion fluxes to the PFCs are expected in the divertor region. The steady-state values of particle fluxes during H-mode scenarios of ITER-scale devices can reach the value up to $10^{23} - 10^{24} \text{ m}^{-2} \text{ s}^{-1}$ along with thermal loads of $\sim 10 \text{ MW m}^{-2}$ order [1,2]. Operation in such an environment imposes strict requirements on the design of PFCs, their regular diagnostics, and, considering the use of deuterium-tritium (D-T) mixture, the fuel retention in them. Besides stationary exposure, PFCs can be subjected to periodic impulse loads during fast transient events, such as edge-localised modes (ELMs).

ELMs are naturally present in H-mode plasmas and mainly originate within the pedestal region [3–6]. The development of the ELM instabil-

ity results in rapid bursts of hot plasma from the confined region into the scrape-off layer (SOL), consequently leading to high-power short-time ($\leq 1 \text{ ms}$) irradiation of PFCs. The latter can drastically affect the behaviour and the lifetime of armour components. Earlier studies [7–11] indicate that heat loads of $1\text{--}10 \text{ GW m}^{-2}$, corresponding to unmitigated type I ELMs in ITER, can induce surface cracking and melting. Bombardment of PFCs with high-energy ions during ELMs can lead to the contamination of the core plasma with wall materials [12–14]. A potentially enhanced erosion is strongly undesired, especially for the case of high-Z materials such as tungsten (W), which is considered as one of the primary candidates for the divertor and the first-wall components [1,15–17].

Another point of concern is the effect of repetitive transient loads on the dynamics of fuel retention in PFCs. Based on the experience from present tokamaks (JET [18], DIII-D [19], EAST [20], etc.), the

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development of ELMs can result in periodic fluctuations of the wall recycling coefficient, which is a ratio of returning particle flux from a surface to the incident particle flux from plasma. The recycling coefficient can increase during the rise of an ELM load because of an enhanced desorption and decrease at the end due to filling of emptied sites in the material. Additionally, some efforts were devoted to simulate the hydrogen retention under ELM-like conditions using transient heat loads (e.g., induced by laser pulses [21,22]), impulse plasma irradiation [23–27] or numerical modelling [28–32]. However, a major number of laboratory studies mainly focused on single or low-frequency ELM-like events, with total irradiation times incomparably lower than expected discharge durations in ITER [1]. Also, the obtained results do not allow for a definitive statement on how transient loads during ELMs affect the evolution of fuel content in PFCs, since both enhancement effects (e.g., [22,24]) and reduction effects (e.g., [23,29]) were declared.

This work intends to broaden the knowledge of the ELM impact on the fuel retention in PFCs utilising one-dimensional reaction-diffusion simulations. As a representative example, we study the dynamics of D trapping in W under combined stationary and transient loads relevant to those expected in ITER. Applying the free-streaming model (FSM) of ELM parallel transport [33,34], the effect of transient loads is analysed for varied ELM properties, trap parameters, and the D recombination rate within three simulation scenarios characterised by different steady-state exposure conditions. Furthermore, we present analytical expressions for the ELM-average D distribution in W under the steady-state approximation.

The remainder of the paper has the following structure. Section 2 describes the applied numerical model and the input parameters for simulation scenarios. Computational and analytical results are presented in Section 3 and Section 4, respectively. The obtained results are briefly summarised in Section 5.

2. Model description

The current section provides details on the reaction-diffusion model, similar to those used in earlier studies on the D retention under ELMs [28–31]. Definitions of stationary and transient loads are declared in corresponding subsections. The equations are numerically solved with the FESTIM code (v.1.2) [35,36] for a simplified 1D geometry representing the shortest distance from the plasma-facing surface to the water-cooling system of ITER divertor monoblocks. Several additional simplifications are made and summarised below together with the inputs for the modelling.

2.1. Geometry

The ITER divertor will be fabricated of composite monoblocks, consisting of W armour with a minimal thickness of 6–8 mm, cylindrical copper (Cu) interlayer with thickness of 1 mm, and CuCrZr copper alloy cooling pipe with an inner diameter of 12 mm and thickness of 1.5 mm (see Fig. 1)). To qualitatively analyse the impact of transient events on the D retention, we focus on the 1D approximation of the thinnest monoblock region between the plasma-facing and the water-cooled sides (highlighted with the red line on the radial axis in Fig. 1 (a)). The resulting geometry slab is shown in Fig. 1 (b) and includes a six-millimetre W layer, a one-millimetre Cu layer, and a one and a half millimetre CuCrZr layer.

This approach was used in recent research [32] on the impact of the Soret effect on the gas retention in W under ELM-like heat loads. Based on the calculations of the temperature evolution, the approximation was shown to comply reasonably with 2D simulations of W surface heating under ELMs [37]. However, it was also highlighted that the lateral (toroidal) temperature distribution along the monoblock depth can deviate from a uniform depending on the irradiation conditions [38,39].

Such variations in temperature will correspondingly result in a non-homogeneous spatial distribution of fuel in a monoblock.

2.2. Heat transfer

The temperature evolution along the simplified geometry is simulated with a homogeneous heat equation assuming a perfect thermal contact at interfaces:

$$C(T)\rho\frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(\kappa(T)\frac{\partial T}{\partial x} \right), \quad (1)$$

where $C(T)$ [J kg⁻¹ K⁻¹] – specific heat capacity, ρ [kg m⁻³] – material density, $\kappa(T)$ [W m⁻¹ K⁻¹] – thermal conductivity, T [K] – temperature. Temperature dependencies of W heat capacity and thermal conductivity are included according to the work of P. Tolias [40], when the corresponding properties of Cu and CuCrZr are defined using polynomial expressions presented by S. Van den Kerkhof et al. [37]. The initial temperature is set equal to $T_0 = 373$ K.

The material heating is defined by the flux balance at the front W surface:

$$\kappa(T)\frac{\partial T}{\partial x} \Big|_{x=0} = -q_{\text{stat}} - q_{\text{ELM}}(t) + \epsilon_W(T^4 - T_0^4), \quad (2)$$

In the RHS of the equation, the first term (q_{stat} [W m⁻²]) represents the heat flux arriving during the inter-ELM phase, which is assumed time-independent. The second term (q_{ELM} [W m⁻²]) corresponds to pulse-periodic heat loads during ELMs. The final one accounts for the heat losses due to radiation with the constant spectral emissivity of W: $\epsilon_W = 0.4$ [41].

The boundary condition on the rear slab surface imitates the heat losses due to forced convection:

$$\kappa(T)\frac{\partial T}{\partial x} \Big|_{x=L} = q_{\text{loss}}(T), \quad (3)$$

where $L = 8.5$ mm – is the total geometry length. According to the specification [42], the convective cooling is assumed to occur with water flow of 12 m s⁻¹ through a swirl tape with a thickness of 2 mm and twist ratio of 2. The coolant temperature is 373 K and the pressure is 30 bar. The dependence of heat losses on the coolant-side temperature is set with a linear interpolation of data presented by T. Marshall et al. [43].

We must note here that the heat transfer equation is initially solved for the composite geometry. However, the geometry is further reduced to the W domain in order to simplify the multi-material problem of the D transport. To keep the temperature distribution unchanged, temperature dependencies of the heat flux flowing through the W-Cu interface were parametrised depending on the irradiation conditions. The parametrised dependencies were then set on the W rear surface as a boundary condition. Details on the validity of the approximation can be found in [32] and are provided in subsection 3.1.

2.3. D transport

The D transport in a composite geometry represents a multi-material problem which requires ensuring the continuity of chemical potential at interfaces [44,45]. To simplify the analysis, the simulations were done only for the W slab with the total length $L_W = 6$ mm, analogically to [32]. On such time-scales, when D does not reach the W-Cu interface, the obtained results will reproduce those for the composite geometry. According to the previous studies [35,44], it would take thousands of seconds (several ITER-like plasma cycles) for D to reach the rear W side. The domain $[0, L_W]$ was discretised using a non-uniform 750-element mesh. The position of the n -th vertex was defined with geometric progression: $\forall n \in (0, 750) : x_n = x_{n-1} + \Delta x b^{n-1}$, with $\Delta x = 0.1$ nm and $b = 1.0188$.

Following the McNabb and Foster approach [46], dynamics of the D transport in W is determined by density evolutions of mobile (c_m [m⁻³]) and trapped species (c_t [m⁻³]):

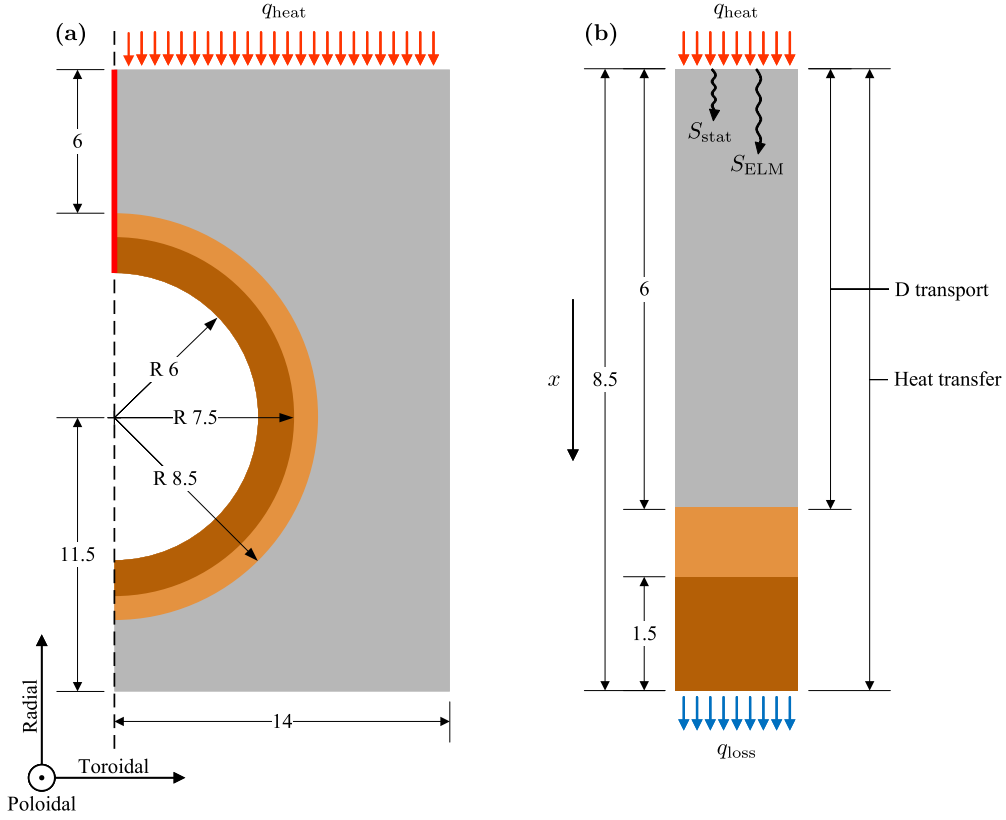


Fig. 1. (a) Poloidal cross-section of an ITER-like W monoblock, considered in the work. Only half is shown because of symmetry. (b) One-dimensional representation of the studied geometry, corresponding to the red line in subfigure (a): ■ - W, ■ - Cu, ■ - CuCrZr. Dimensions are shown in mm. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

$$\frac{\partial c_m}{\partial t} = -\frac{\partial J}{\partial x} - \frac{\partial c_t}{\partial t} + S_{\text{stat}}(x) + S_{\text{ELM}}(x, t); \quad (4a)$$

$$\frac{\partial c_t}{\partial t} = c_m v_t(T) \left(1 - \frac{c_t}{n_t} \right) - c_t v_{\text{dt}}(T). \quad (4b)$$

In eq. (4a), the D flux (J) is governed by both Fick's law of diffusion and the Soret effect:

$$J = -D(T) \left(\frac{\partial c_m}{\partial x} + \frac{Q(T)c_m}{kT^2} \frac{\partial T}{\partial x} \right). \quad (5)$$

The Soret effect is included since it was shown to affect heavily the D retention in W under ELM-like conditions [32], whereas the effect of stress fields [47] is not considered. The second term in the RHS of eq. (4a) represents the net exchange rate with the trapped population, when the two last terms correspond to implantation sources of mobile species due to stationary and transient particle loads. Diffusion is assumed to occur via interstitial sites with the diffusion coefficient expressed in the Arrhenius form: $D = D_0 \exp(-E_d/kT)$, where $D_0 = 1.93 \times 10^{-7}/\sqrt{2} \text{ m}^2 \text{ s}^{-1}$ – diffusion pre-factor, $E_d = 0.2 \text{ eV}$ – diffusion activation energy, and k – the Boltzmann constant. The diffusion coefficient is chosen according to the results of DFT calculations [48] and scaled by $1/\sqrt{2}$ to apply it for the D case. The Soret coefficient is taken from [32] and expressed as: $Q(T) = -0.0045 kT^2 \text{ [eV]}$. The initial concentration of mobile species is set equal zero.

Volumetric sources of mobile atoms (in $\text{m}^{-3} \text{ s}^{-1}$), S_{stat} and S_{ELM} , refer to implantation of low-energy D during stationary loads and high-energy D during transient events. They are given with:

$$S_i(x, t) = \Gamma_i^{\text{imp}}(t) \varphi_i(x), \quad (6)$$

where $\Gamma_i^{\text{imp}} = \Gamma_i(1 - r_i) \text{ [m}^{-2} \text{ s}^{-1}]$ is the stationary/transient implantation D flux, r_i is the particle reflection coefficient from a W surface.

Spatial distribution of each implantation source (φ_i) is approximated with a normalised Gauss function:

$$\varphi_i(x) = \frac{N_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x - X_i)^2}{2\sigma_i^2}\right); \quad (7a)$$

$$N_i = \frac{2}{\text{erf}\left(\frac{L - X_i}{\sqrt{2}\sigma_i}\right) + \text{erf}\left(\frac{X_i}{\sqrt{2}\sigma_i}\right)}, \quad (7b)$$

where $\text{erf}(x)$ is the Error function, $X_i \text{ [m]}$ and $\sigma_i \text{ [m]}$ are the mean implantation depth and the standard deviation of the distribution, respectively. The particle reflection coefficient and distribution parameters are assumed time-independent. For each case, they were extracted from statistics on implanted/backscattered atoms generated with SDTrimSP (v6.03) [49]. The statistics (10^8 projectiles in total) was obtained in the static regime for the normal incidence of impinging D ions.

The evolution of trapped species is determined by the interplay between trapping and detrapping processes. The first term in the RHS of eq. (4b) defines trapping of D into saturable immobile sites of one kind with concentration $n_t \text{ [m}^{-3}]$. One type of traps is used to simplify the parametric study, but a more general model would require setting at least two trapping sites to characterise intrinsic defects for D in W [50,51]. The trapping attempt frequency ($\nu_t \text{ [s}^{-1}]$) is assumed to follow the Arrhenius law: $\nu_t(T) = n_t \nu_{t,0} \exp(-E_t/kT)/n_{\text{IS}}$, where $n_{\text{IS}} = 6 n_W$ [48] is the concentration of tetrahedral interstitial sites in W and $n_W \approx 6.3 \times 10^{28} \text{ m}^{-3}$ is the W atomic density. Further in the paper, the amount of D traps in W is characterised by atomic fraction: $\eta_t = n_t/n_W$. Trapping is assumed to be diffusion-limited: $E_t \approx E_d$ and $\nu_{t,0} \approx D_0/\lambda^2$ with $\lambda = 1.1 \times 10^{-10} \text{ m}$ being the characteristic jumping distance between two interstitial sites in W [48].

The second term characterises detrapping from traps with the attempt frequency $\nu_{\text{dt}} \text{ [s}^{-1}]$. It is expressed with: $\nu_{\text{dt}}(T) = \nu_{\text{dt},0} \exp(-E_{\text{dt}}/$

Table 1
Stationary exposure conditions.

$q_{\text{stat}}, \text{MW m}^{-2}$	$\Gamma_{\text{stat}}, 10^{23} \text{ m}^{-2} \text{ s}^{-1}$	$T_{\text{e,t}}, \text{eV}$	$E_{\text{stat}}, \text{eV}$	r_{stat}	$r_{\text{stat}}^{\text{E}}$	$X_{\text{stat}}, \text{nm}$	$\sigma_{\text{stat}}, \text{nm}$
1	1.0	10.5	51	0.685	0.484	2.64	1.42
5	3.6	15.5	75	0.666	0.461	3.27	1.76
10	5.7	20.0	97	0.654	0.447	3.76	2.03

kT), where $\nu_{\text{dt},0} = 10^{13} \text{ s}^{-1}$ – pre-exponential frequency factor [48], E_{dt} [eV] – detrapping activation energy. The traps are assumed to be uniformly distributed in W with a zero initial filling ratio. The possible contribution from ion-induced traps [52] and supersaturated layer [53] is neglected because long-time D retention is expected to be determined by the transport in the bulk [54].

To analyse the impact of trap properties on the D retention, the detrapping energy was varied in the range from 1 eV to 2 eV at the fixed trap concentration of 10^{-2} at.%, while the trap concentration – in the range from 10^{-3} at.% to 1 at.% at the fixed detrapping energy of 1.5 eV. In other cases, $E_{\text{dt}} = 1.5 \text{ eV}$ and $\eta_{\text{t}} = 10^{-2}$ at.% were used as the base combination that can be attributed to the natural concentration of vacancy-type defects in W [55].

Concerning boundary conditions, the release of D from material under fusion-relevant conditions is often assumed to be diffusion-limited, i.e. the characteristic time of D recombination and desorption from a W surface is much lower than the time required for D to reach this surface [30,56,57]. At high material temperatures, surface processes governing desorption can affect the D retention on time-scales significantly smaller than the typical duration of a single ELM burst [58]. Therefore, most of the calculations were done within the approximation of the instantaneous desorption imposing homogeneous boundary conditions on both sides:

$$c|_{x=0} = c|_{x=L_{\text{W}}} = 0. \quad (8)$$

A part of the work is dedicated to the influence of the desorption rate on the D retention. For this purpose, a recombination flux is set on the front surface with the recombination coefficient K_{r} [$\text{m}^4 \text{ s}^{-1}$] according to [59]:

$$J|_{x=0} = -K_{\text{r}}(T)c_{\text{m}}^2; \quad (9a)$$

$$K_{\text{r}}(T) = \frac{K_{\text{r},0}}{\sqrt{T}} \exp\left(2\frac{Q_{\text{s}} - E_{\text{c}}}{kT}\right), \quad (9b)$$

where $K_{\text{r},0} = 3 \times 10^{-25} \text{ m}^4 \text{ s}^{-1} \text{ K}^{1/2}$, $Q_{\text{s}} = 1.03 \text{ eV}$ is the heat of solution, E_{c} [eV] is the chemisorption activation energy. The effect of the recombination rate was analysed by varying the chemisorption activation energy in the range from 0 to 0.83 eV at the fixed trap properties. The lower range value corresponds to the case of a “clean” W surface, whereas non-zero ones can be attributed to an existence of impurities, limiting the D release [59], or any other reasons.

2.4. Stationary loads

Influence of ELMs on the D retention in W is studied for three inter-ELM exposure conditions. These conditions are derived, implying the ITER baseline scenario at $Q = 10$ (15 MA/5.3 T), though assuming a pure D plasma. Particularly, the focus is made on the irradiation conditions expected at the outer divertor target. The next subsection will additionally highlight that the approximation of transient loads is applicable for the outer divertor target. Stationary heat loads were chosen equal to 1, 5, and 10 MW m^{-2} . According to the SOLPS simulations [1,2], such background heat fluxes can be expected near the strike point in attached and semi-detached regimes.

The steady-state particle flux for each case is expressed via heat power density and electron temperature at the divertor target ($T_{\text{e,t}}$ [eV]), assuming thermal equilibrium between ion and electron plasma components [60]:

$$\Gamma_{\text{stat}} = \frac{q_{\text{stat}}}{\gamma T_{\text{e,t}} + E_{\text{rec}}}; \quad (10a)$$

$$\gamma = 4.85(1 - r_{\text{stat}}^{\text{E}}) + 2.15, \quad (10b)$$

where γ – sheath heat transmission factor, $r_{\text{stat}}^{\text{E}}$ – energy reflection coefficient of low-energy D impinging the target, $E_{\text{rec}} = 13.6 \text{ eV}$ – energy deposited at the target due to ion recombination. Eqs. (10) accounts for sheath effects, neutralisation of ions near W surface, and reflection of ions. Since D ions experience additional acceleration by the sheath potential, their kinetic energy at the target is approximately [60]:

$$E_{\text{stat}} \approx 4.85 T_{\text{e,t}}. \quad (11)$$

To obtain the values of D fluxes and incident energies, electron temperatures at the target were chosen to be 10.5, 15.5, and 20.0 eV, consistent with the results of laborious edge plasma simulations [1,2]. The parameters of stationary loads are summarised in Table 1 and generally close to those applied in other reaction-diffusion simulations of hydrogen retention in ITER divertor components [35,38,39,44,54].

2.5. Transient loads

Time-dependent heat and particle loads during ELMs are defined according to the free-streaming model (FSM), comprehensively described in [33,34]. Within the approximation of quasi-neutrality, FSM solves the Vlasov-Poisson equations for collisionless adiabatic expansion of the plasma filament in SOL after detachment from the pedestal region. It is widely applied for the investigation of W gross erosion [12,61–65], fuel retention [32] and thermal behaviour of PFCs [37,66] under ELM-like conditions in tokamaks.

The resulting temporal dependencies of heat (q_{ELM}) and particle (Γ_{ELM}) fluxes at the target can be expressed as:

$$\frac{q_{\text{ELM}}(t)}{q_{\text{ELM},0}} = \left(1 + \left(\frac{\tau}{t}\right)^2\right) \left(\frac{\tau}{t}\right)^2 \exp\left(-\left(\frac{\tau}{t}\right)^2\right); \quad (12a)$$

$$\frac{\Gamma_{\text{ELM}}(t)}{\Gamma_{\text{ELM},0}} = \left(\frac{\tau}{t}\right)^2 \exp\left(-\left(\frac{\tau}{t}\right)^2\right), \quad (12b)$$

where $\tau = 0.8 \tau_{\text{IR}}$ with $\tau_{\text{IR}} = 250 \mu\text{s}$ being the time-constant of the ELM rise phase [67]. The normalisation factor $q_{\text{ELM},0}$ is derived from the spatial peak energy fluence of the ELM ($\epsilon_{\parallel}$ [J m^{-2}]) following the approach described in [37]:

$$\epsilon_{\parallel} = \sin^{-1}(\alpha) \int_{\tau_{\text{ELM}}} q_{\text{ELM}}(t) dt, \quad (13)$$

where τ_{ELM} is the ELM period (reciprocal of the ELM frequency), $\alpha = 4.2^\circ$ is the total outer target field line angle [1,67,68]. The ELM parallel energy fluence at the outer target is obtained with the mean-value scaling, provided by T. Eich et al. [67]:

$$\epsilon_{\parallel} = 0.28 \times n_{\text{e,ped}}^{0.75} T_{\text{e,ped}}^{0.98} \delta_{\text{ELM}}^{0.52} R_{\text{geo}}^1, \quad (14)$$

where $\epsilon_{\parallel}$ is expressed in MJ m^{-2} , $n_{\text{e,ped}}$ [10^{20} m^{-3}] is the electron pedestal density, $T_{\text{e,ped}}$ [keV] is the electron pedestal temperature, $\delta_{\text{ELM}} = \Delta W_{\text{ELM}}/W_{\text{plasma}}$ [%] is the ELM energy loss relative to the plasma stored energy W_{plasma} [J], R_{geo} [m] is the geometrical major radius. The ELM energy loss is given by empirical relation with the power flowing into SOL (P_{SOL}) [69]: $\Delta W_{\text{ELM}} f_{\text{ELM}} = 0.3 \pm 0.1 P_{\text{SOL}}$, where f_{ELM} is the ELM frequency. As for the ELM energy density, we focus

Table 2

Parameters of transient loads computed with $E_{\text{ELM}} = 9.4 \text{ keV}$, $r_{\text{ELM}}^E = 0.139$.

f_{ELM} , Hz	$q_{\text{ELM},0}$, GW m^{-2}	$\Gamma_{\text{ELM},0}$, $10^{23} \text{ m}^{-2} \text{ s}^{-1}$
10	1.70	26.35
20	1.19	18.41
30	0.97	14.93
40	0.83	12.87
50	0.74	11.48
60	0.68	10.46
70	0.63	9.67
80	0.58	9.03
90	0.55	8.51
100	0.52	8.07

on the mean value of the relation, namely set the ELM power equal to 30% of the SOL power.

Similarly to the stationary case, the ELM particle flux at the target is expressed via the ELM heat flux. To obtain $\Gamma_{\text{ELM},0}$, one can use the representation of the heat flux presented in [33,34], modified to account for a reduction in the energy deposition due to particle reflection:

$$q_{\text{ELM}}(t) = \Gamma_{\text{ELM}}(t) T_{e,\text{ped}} (1 - r_{\text{ELM}}^E) \left(1 + \left(\frac{\tau}{t} \right)^2 \right). \quad (15)$$

Substitution of eq. (12b) into eq. (15) and comparison with eq. (12a) yield:

$$\Gamma_{\text{ELM},0} = \frac{q_{\text{ELM},0}}{T_{e,\text{ped}}(1 - r_{\text{ELM}}^E)}. \quad (16)$$

Transient loads are assumed to reach the target at the same point as stationary loads, but both can have a broad distribution over the outer target depending on the conditions [1,2,67]. Besides, ELM filaments can also reach the inner divertor target as well as PFCs of the main chamber due to radial propagation [6]. The expressions presented above are supposed to hold for mitigated ELMs with frequencies of $\sim 10 \text{ Hz}$. For the calculations, we set $n_{e,\text{ped}} = 8 \times 10^{19} \text{ m}^{-3}$, $T_{e,\text{ped}} = 4.7 \text{ keV}$, $W_{\text{plasma}} = 350 \text{ MJ}$, $P_{\text{SOL}} = 100 \text{ MW}$, $R_{\text{geo}} = 6.2 \text{ m}$, corresponding to the ITER baseline operational scenario at $Q = 10$, and vary the ELM frequency in the range from 10 Hz to 100 Hz. A possible time-variation in plasma parameters during ELMs (e.g., electron pedestal density and temperature drop) is not considered herein, but possible approaches to account for it can be found elsewhere [62,63].

To set the ELM source of implanted D in eq. (4a), a value of ion target energy is required. However, a precise definition of ion target energy during transients is an open question. Based on the indirect estimations of electron target energy using coupled infrared thermography and Langmuir probe measurements in JET [70,71], it was concluded that electron transfer more of their parallel energy to ions during the ELM filament transport. Low calculated electron temperatures in the divertor allowed to derive the following relation between the peak ion energy and the pedestal electron temperature: $E_{\text{ELM}}^{\text{max}} = 4.23 T_{e,\text{ped}}$. More recent measurements from COMPASS [72] and the re-analysis of the JET data [73] though report higher electron temperatures in the divertor region during ELMs, suggesting an absence of significant acceleration of ions during transport in the SOL.

In view of the described uncertainty, a simplified representation of the ELM source was used. For most simulations, implantation and reflection parameters were calculated assuming monoenergetic D ions arriving at the W surface with $E_{\text{ELM}} = 2 T_{e,\text{ped}}$ at the normal incidence angle. The calculated parameters and resulting temporal dependencies of heat and particle loads are presented in Table 2 and illustrated in Fig. 2. An exception is made in Subsection 3.3, where the impact of the ELM particle flux-to-energy ratio ($\Gamma_{\text{ELM}}/E_{\text{ELM}}$) on the D retention is analysed. For that part of work, the particle flux during ELMs is scaled by a factor of β and the ion energy - by a factor of $1/\beta$, preserving the heat power density. That way, the particle flux-to-energy ratio is increased by a factor

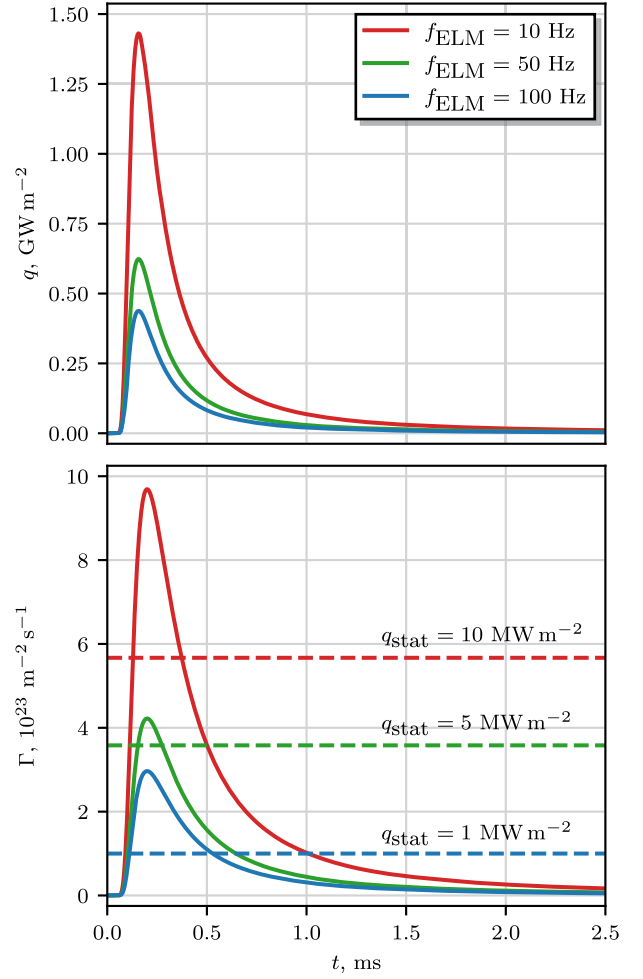


Fig. 2. Temporal dependencies of heat and particle fluxes during ELMs. Dashed lines correspond to inter-ELM particle fluxes under different heat loads.

Table 3

Implantation and reflection parameters calculated for different energies of ELM ions.

β	E , eV	r_{ELM}	r_{ELM}^E	X_{ELM} , nm	σ_{ELM} , nm
1	9400	0.298	0.139	73.4	38.2
1.33	7050	0.334	0.164	58.8	31.0
2	4700	0.382	0.200	43.4	23.2
2.5	3760	0.407	0.218	36.9	19.8
5	1880	0.474	0.273	22.7	12.3
10	940	0.529	0.322	14.4	7.8

of β^2 . Implantation and reflection parameters were computed for each case and are presented in Table 3. Details on the choice of the range for β are provided in the corresponding subsection. Possible mechanisms of surface and bulk modification due to high-energy ion bombardment are not considered for the sake of simplicity.

3. Results of numerical calculations

This section summarises the results of numerical calculations. It begins with a brief illustration of material response to high-power heat loads during ELMs and provides support for the geometry reduction made in the study. Subsection 3.2 and Subsection 3.3 are devoted to the influence of transient exposure conditions. The final two subsections include results of the trap parameter scan (Subsection 3.4) and the recombination rate variation (Subsection 3.5) that can be attributed to the impact of material properties on the D retention under ELM-like con-

ditions. To demonstrate the general trends, time-dependent simulations were carried out for 1000 s of continuous plasma exposure.

3.1. Temperature evolution

It is well known that material temperature is a crucial factor determining the retention of implanted hydrogen [74]. After the onset of ELMs, periodic oscillations of the near-surface temperature are expected due to a repetitive deposition of additional heat. The amplitude of fluctuations depends on the heat flux factor ($\propto \varepsilon_{\parallel} \sin(\alpha) / \sqrt{\tau_{\text{ELM}}}$) of the ELM load. Large unmitigated ($f_{\text{ELM}} \sim 1$ Hz) ELMs, carrying a significant amount of the plasma stored energy, can easily induce melting of a surface or even obliterate it. Therefore, it is essential to ensure that the parameters of the introduced above model correspond to the undamaging heating regime. The melting temperature can be used as a rough threshold.

Dynamics of the surface temperature at $q_{\text{stat}} = 10 \text{ MW m}^{-2}$ is shown in Figs. 3 (a)–(b) for several ELM frequencies. Time-dependencies quickly reach the quasi-plateau within a time period of ≈ 3 s, much shorter than the expected discharge duration (~ 100 s) in ITER. The quasi-plateau is characterised by ELM-induced oscillations of temperature, which does not exceed the W melting point (≈ 3695 K). Within the described model, transient loads with lower frequencies correspond to higher temperature fluctuations (see Fig. 3 (b)), but the ELM-averaged surface temperature rises oppositely as shown with dashed lines. Oscillation amplitudes correlate with the results of 2D simulations [37], but ELM-averaged temperatures are underestimated due to the use of one-dimensional approximation [32].

As was initially shown in [32], the spatial variations of temperature are limited to the W domain after the onset of the quasi-plateau. To demonstrate the veracity of the proposal under different exposure conditions, shaded areas in Fig. 3 (c) define the calculated temperature variations over one ELM period, accompanied by temperature distributions with the highest/lowest surface values. It is clear that ELMs induce temperature disturbances only near the W surface, whereas the deeper part of the composite geometry remains within the steady-state regime. Moreover, the size of the disturbed zone decreases with the ELM frequency. This provides an opportunity to reduce the formulation of the heat transfer problem to the W slab. The reduction was made by parametrising the heat flux at the W-Cu interface:

$$q_{\text{loss}}^* = \bar{q}_{\text{tot}} \left(a + b(T - T_0) - c \cdot \exp\left(-\frac{T - T_0}{d}\right) \right); \quad (17a)$$

$$\bar{q}_{\text{tot}} = q_{\text{stat}} + f_{\text{ELM}} \int_{\tau_{\text{ELM}}} q_{\text{ELM}}(t) dt. \quad (17b)$$

The second term in eq. (17b) represents the ELM-average power density and, taking into account eq. (13), simply equals to $\bar{q}_{\text{ELM}} = f_{\text{ELM}} \varepsilon_{\parallel} \sin(\alpha)$. With fitting parameters listed in Table A.1, these expressions were then used in a boundary condition for the heat transfer equation, enforced on the rear W surface.

3.2. Effect of the ELM frequency

The evolution of implanted D in W under combined (stationary and transient) exposure was analysed depending on the ELM frequency and the background irradiation conditions at $E_{\text{dt}} = 1.5$ eV, $\eta_t = 10^{-2}$ at.%. With solid lines, Fig. 4 shows temporal dependencies of surface temperature and D content in W ($\int (c_m + c_t) dx$) for 1000 s of continuous irradiation within three simulations scenarios, whereas dashed lines correspond to the case of pure stationary exposure. To clearly demonstrate the tendency, only ELM-averaged dependencies are presented. Besides, the colour definition simply states that the darker the colour, the higher the ELM frequency.

At early stages ($t < 1$ s), combined irradiation of W leads to an increase in the amount of D retained in W compared to the stationary

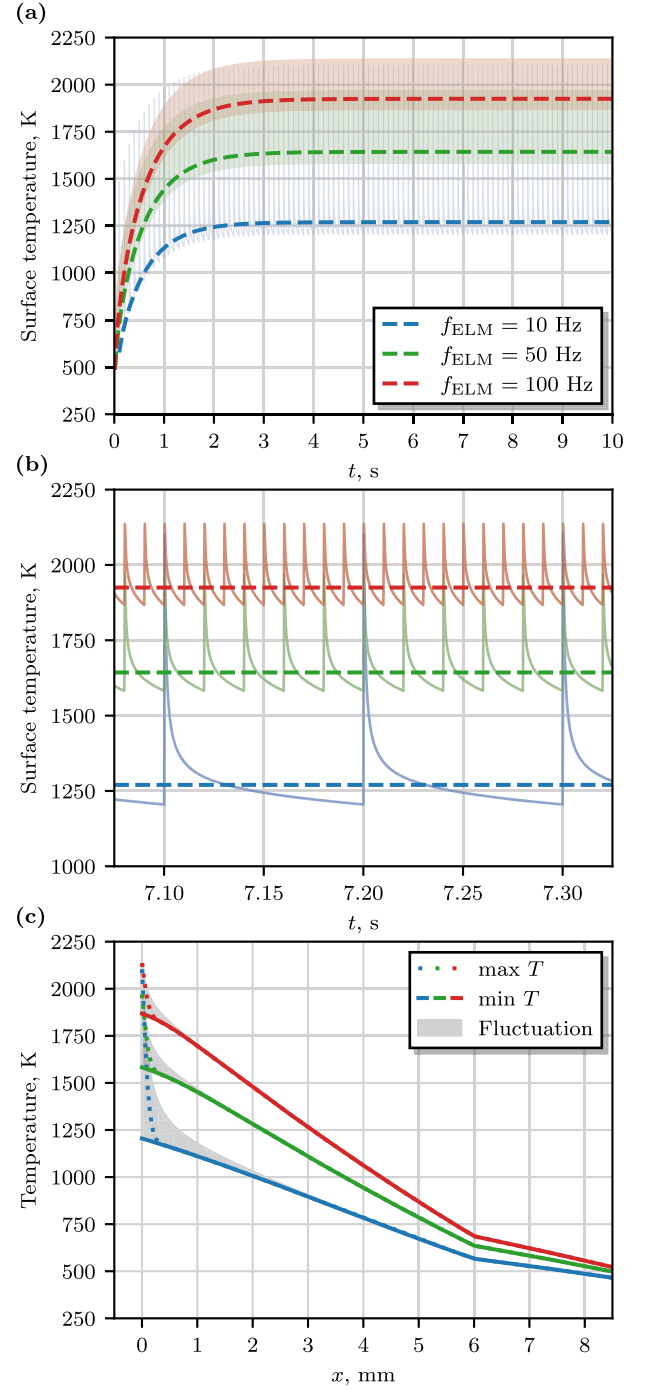


Fig. 3. (a) Temporal evolution of the front surface temperature under combined heating with $q_{\text{stat}} = 10 \text{ MW m}^{-2}$ and different ELM frequencies. Dashed lines show dependencies averaged over the ELM duration. Shaded areas are individual temperature spikes clearly shown with a shorter time slice in figure (b). (c) Region of temperature change along the composite geometry over one ELM period from figure (b). Solid and dotted lines show depth distributions when maximal and minimal surface temperatures are reached. Colour definition in figure (a) is the same for all plots.

exposure only. The increase can be easily understood from the maximal concentration of mobile species within the approximation of fast desorption [30,38]:

$$c_m^{\text{max}} \propto \frac{\Gamma X}{D(T)}. \quad (18)$$

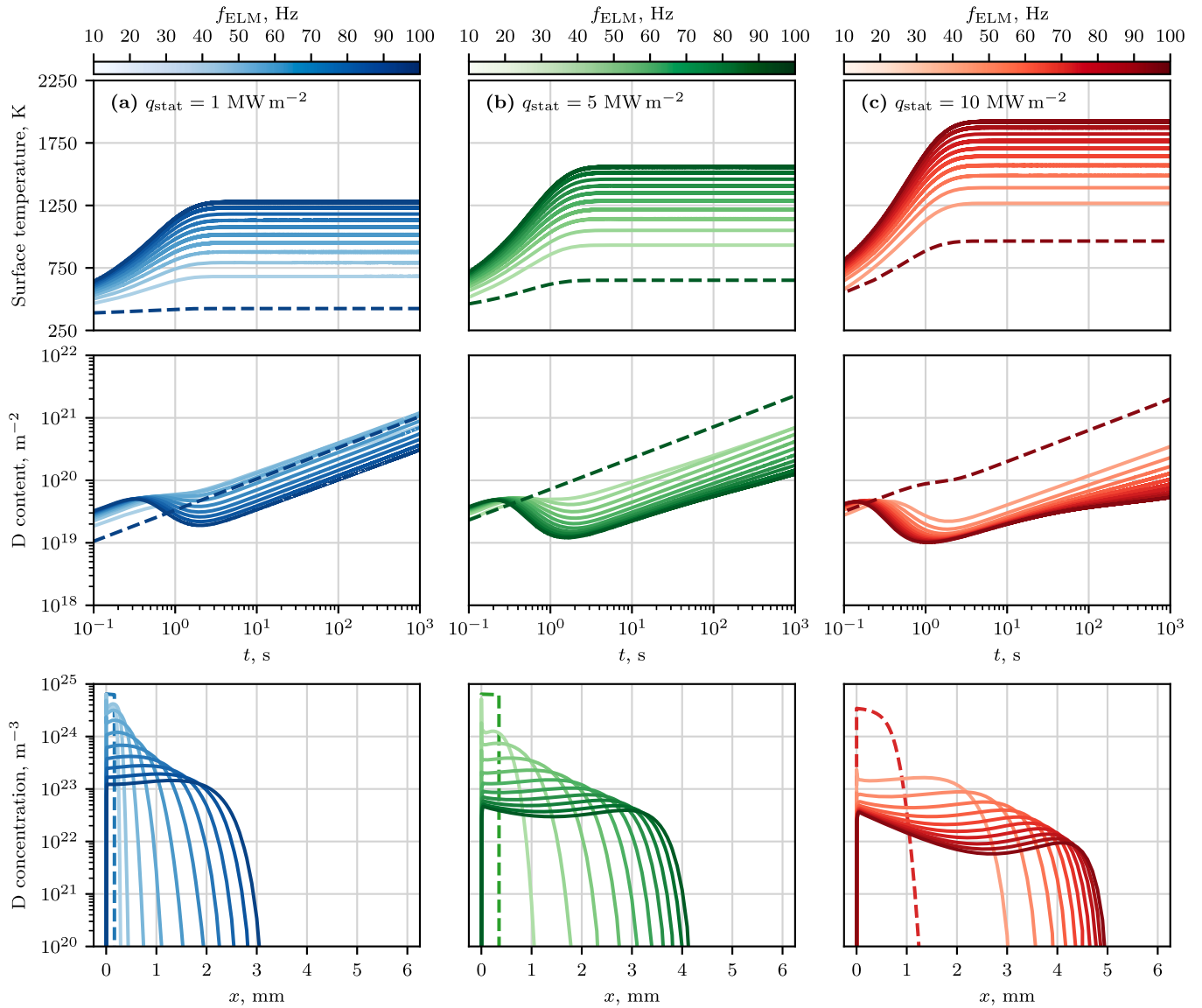


Fig. 4. Temporal dependencies of ELM-averaged surface temperature (top row) and D content (middle row) under different stationary loads at $E_{\text{dt}} = 1.5 \text{ eV}$, $\eta_i = 10^{-2}$ at.%, $F_{\text{ELM}} = 2 T_{\text{e,pqd}}$: column (a) - $q_{\text{stat}} = 1 \text{ MW m}^{-2}$, column (b) - $q_{\text{stat}} = 5 \text{ MW m}^{-2}$, column (c) - $q_{\text{stat}} = 10 \text{ MW m}^{-2}$. Dashed lines show the corresponding dependencies under stationary loads only. The bottom row depicts the D depth distributions at the final time-step.

At low temperatures, an additional flux can raise the value of $c_{\text{m}}^{\text{max}}$. Since ELM heat and particle loads are identical in all scenarios, the increase is more apparent at $q_{\text{stat}} = 1 \text{ MW m}^{-2}$ (see column (a) in Fig. 4), where the background loads are the lowest. In this case, the enhancement in D retention also remains for higher irradiation fluences at low ELM frequencies, i.e. when the average induced temperature rise and the desorption rate are not high. These observations can be speculated to correlate with experimental results [24–26] where a similar increase in D retention under ELM-like loads was observed either with a minor number of plasma shots or with long resting times between pulses.

On the other hand, most of the dependencies exhibit a decline after the onset of the high-temperature quasi-plateau ($t > 1 \text{ s}$). The drop in the D content increases with the ELM frequency and the stationary load due to higher base temperatures. Additional heating during ELMs raises the average temperature by a minimum of $\sim 260 \text{ K}$ at (10 Hz, 1 MW m^{-2}) and by a maximum of $\sim 1000 \text{ K}$ at (100 Hz, 10 MW m^{-2}) compared to the pure background exposure. In general, this temperature rise is induced by the arrival of high-energy ($E \sim T_{\text{e,pqd}}$) particles during ELMs. However, the average flux of D ions is close to the background one. Thus, the observed resulting effect of ELM-like loads is a

reduction of the total D content in W, which is consistent with earlier research [29]. Meanwhile, the highest retention is reached in the intermediate stationary case with $q_{\text{stat}} = 5 \text{ MW m}^{-2}$.

Overall, the appearance of transient events mostly reduces the rate of D accumulation in W due to an establishment of a higher-temperature regime. Higher material temperatures provide conditions for an enhanced D diffusion into the bulk. To illustrate, the bottom row in Fig. 4 presents depth distributions of total D concentration at $t = 1000 \text{ s}$. From the figure, it is evident that D migrates far deeper into the bulk under combined irradiation. It is also important to highlight that this observation is valid in all other cases shown further. Under stationary conditions, similar migration depths of millimetres can be reached on significantly larger time-scales [54].

Regarding D depth profiles, one can notice several qualitative trends. Firstly, the peak values of D concentration are located near the front surface due to a small implantation depth and a narrow implantation profile compared to the domain size. Secondly, broader D distributions at higher average temperatures have lower peak values in comparison with narrow ones. Finally, D distributions tend to rise near the rear W side, where the temperature is lower. The last two observations can be

explained by the characteristic migration rate within a certain heating regime and the equilibrium trap filling ratio, which tends to decrease with temperature [75].

The obtained results allow to suppose that long-time transient loads due to mitigated ELMs can potentially reduce fuel accumulation rate in a PFC. Moreover, harmless ELMs can similarly affect the fuel content in surface layers formed due to co-deposition, which is considered being one of the main mechanisms for long-term fuel retention in a tokamak environment [76,77]. In particular, this effect can provide a benefit with regard to the tritium control. At the same time, enhanced diffusivity during transient events can shorten the so-called “breakthrough” time for tritium reaching the water-cooling system of the divertor monoblocks.

3.3. Effect of the ELM particle flux-to-energy ratio

The previous subsection demonstrated that additional periodic irradiation with high-energy particles can reduce the rate of D accumulation in W if the transient particle flux is comparable to the background one. The ion flux during ELMs can, though, change near the target due to secondary effects, not covered by the FSM equations (12). These expressions were derived assuming collisionless transport of a hot plasma filament from the pedestal region to perfectly absorbing divertor targets in the absence of a background plasma. Therefore, some processes, e.g. interaction of ELM particles with each other and/or the background ones, can alter the target exposure conditions.

Earlier findings [34,78] predict that particle collisions during the ELM filament propagation can affect the target exposure conditions. The main consequence is a pronounced increase in the peak value of the ELM power load due to the conversion of the electron perpendicular energy to the parallel energy of ions and electrons. It is worth noticing that this effect weakens with a decrease in the collisionality, which is expected to be low in ITER. According to subsection 3.2, a disproportionate rise of the transient heat flux will result in a reduction in the D retention rate.

Another group of processes, demanding attention, is related to particle recycling. To account for it, T. Abrams et al. [61] assumed that the ELM particles are rapidly ionised and thermalised in the divertor plasma after being recycled with a certain probability from the target. By following the magnetic field lines, newly created ions again reach the target. This process is repeated multiple times, forming an additional low-energy ion flux. The main difficulty with this approach is the use of an effective recycling probability, which is treated as an empirical parameter.

A more detailed description of possible near-surface mechanisms is given by C. Guillemaut et al. [71]. Following the authors' qualitative discussion, energetic ELM ions can interact with the ones recycled either due to reflection from the target or due to implantation, followed by desorption. Both recycling processes mostly result in the formation of neutral particles: desorbed D_2 molecules and reflected D atoms. Due to interaction with a dense incoming flux of particles, these neutral species can eventually come back to the target, levelling up heat and particle fluxes during ELMs.

For the sake of illustration, we can divide particles into two groups, namely fast, related to ELMs (ions, reflected atoms), and slow, referred to the background plasma (ions, reflected atoms) and desorption (molecules). As was shown [71], the charge-exchange (CX) reaction is a feasible mechanism for the interaction of fast particles (fast D ion - fast D atom) that can raise the particle/heat flux onto the surface (see the schematic diagram in Fig. 5). Similarly, fast ions can interact with slow D atoms formed due to reflections or due to dissociation of D_2 molecules, for example, by electron impact (EI). Through this CX-reaction, fast particles can also transfer a part of their energy to slower ones affecting the implantation process. Lastly, slow atoms can become ions via direct EI. All these processes can take place several times depending on the characteristic particle energy, surface condition, background plasma properties, etc.

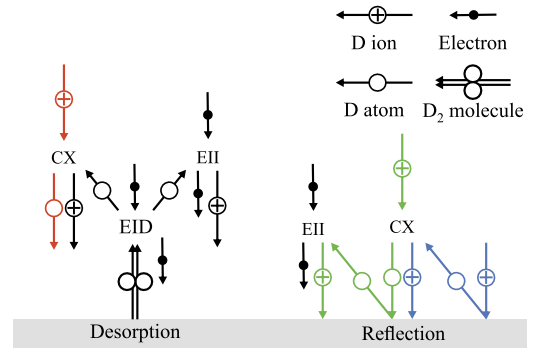


Fig. 5. Schematics of possible processes near a target surface leading to an increase in the flux of particles during ELMs: — target material, CX - charge exchange, EII - electron impact ionisation, EID - electron impact dissociation. Individual particles can be tracked through processes by following the lines of the same colour.

The impact of recycling on hydrogen isotope retention under transient events requires dedicated research. Nonetheless, a simplified case of a flux-to-energy ratio change during transport through a dense near-surface plasma is considered. We assume that some processes during transport result in an instantaneous energy exchange between fast ELM ions and slow near-target particles. On average, this energy transfer can lead to an increase in the ELM particle flux by a factor of β and to a proportional decrease in their impact energy, therefore, the flux-to-energy ratio increases by β^2 . With that, the transient heat flux remains nearly constant. For the present study, β was varied in the range from 1 to 10. The choice of this range was motivated by the work of T. Abrams et al. [61], where a low-energy recycling flux was introduced to better reproduce the experimental measurements. To limit the range of possible values, $\beta = 10$ was chosen as the upper limit, so the minimal ion energy corresponded to 10% of the one in the base simulation case.

Fig. 6 (a) presents the values of D content in W after 1000 s of combined irradiation at different ELM frequencies. An increase in β more pronouncedly raises the retention in low-temperature regimes. It might be seen by comparing the results both at a fixed ELM frequency and at a fixed stationary load. Although the observed is an expected outcome, it indicates a possible comparability of the D content values reached under combined and stationary only exposure (see the data for $q_{\text{stat}} = 1 \text{ MW m}^{-2}$). Qualitatively, the enhancement in the D retention rate can be understood from eq. (18). Based on Table 3, an increase in β raises moderately the product $\beta \Gamma_X(E/\beta)$ and, consequently, the overall retention. Therefore, some enhancement in the retention rate during transients can be expected in the case of high-flux irradiation with low-energy ions.

3.4. Effect of D trap properties

Parameters of traps determine the probabilities of D to get caught in a defect site and to leave it. From eq. (4b), it is obvious that once D is trapped, it will more likely stay in high-energy traps (i.e. trapping sites with a high detrapping barrier) during irradiation rather than in low-energy ones. It is also trivial to conclude that damaged materials with high concentration of traps are expected to retain more of implanted D than those of ideal structure. However, the choice of specific trap parameters for the simulations is an open question due to an uncertainty related to the manufacturing process of PFCs. Even though the initial values of trap properties can be limited to a particular range, they can eventually change because of intense plasma and neutron exposure during operations of a fusion device.

The mentioned above effects can be seen in rows (b) and (c) of Fig. 6, where the simulation results are shown for sequentially varied trap concentration and detrapping energy. In all simulation scenarios,

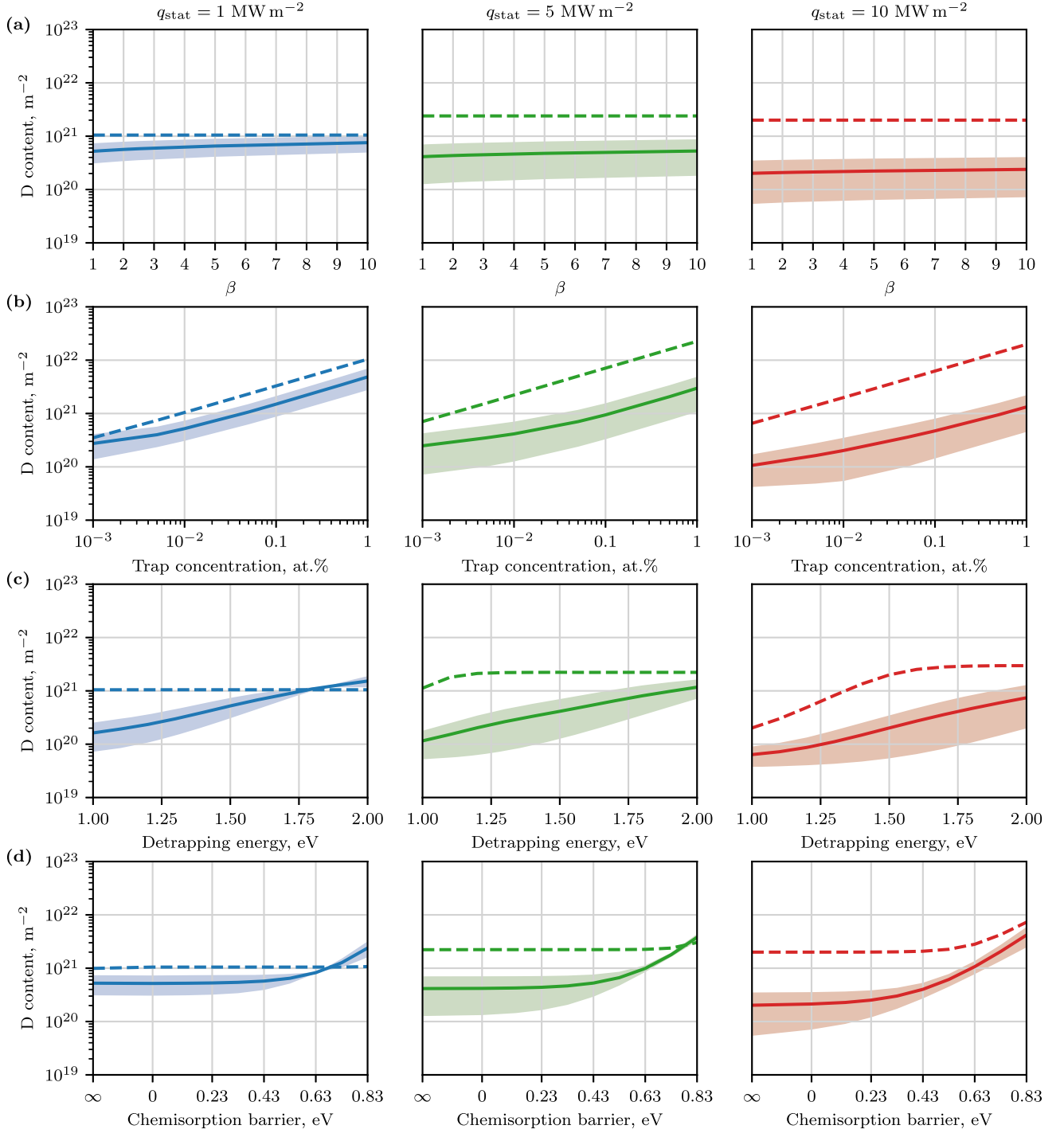


Fig. 6. Dependencies of the D retention after 1000 s of continuous irradiation: (a) - at different values of the ELM particle flux; (b) - at different values of the detrapping energies from traps, (c) - at different trap concentrations, (d) - at different rates of D recombination on a surface. The parameters are varied from the base case: $E_{\text{ELM}} = 2T_{\text{e,ped}}$, $E_{\text{dt}} = 1.5$ eV, $\eta_t = 10^{-2}$ at.%. The results in plots (a)-(c) were obtained within the assumption of an instantaneous recombination, which is also shown in panel (d) with the infinity symbol. Solid lines of each colour represent the mean values obtained under combined irradiation with different ELM frequencies. Top bounds of shaded areas correspond to $f_{\text{ELM}} = 10$ Hz, bottom - $f_{\text{ELM}} = 100$ Hz. Dashed lines show the data obtained under stationary exposure only.

the D retention increases with the trap concentration at the fixed detrapping energy of 1.5 eV (see row (b) of Fig. 6), suggesting a higher retention rate during exposure. For the fixed implantation conditions, this observation means that a larger fraction of implanted D atoms are being trapped in defect sites during diffusion. However, the reduction in the D content under combined irradiation is still observed for the used set of parameters and it is more evident in the case of traps with high concentration.

The effect of the detrapping barrier (E_{dt}) on the D retention is shown in row (c) of Fig. 6. In low-temperature regimes (e.g. stationary exposure only at $q_{\text{stat}} = 1 \text{ MW m}^{-2}$), there is no difference between high-energy and low-energy traps in terms of the retention, since the major fraction of implanted D is being trapped in defect sites. In high-temperature regimes, the D content increases with the detrapping barrier, as a larger fraction of D atoms can remain in the trapped state during exposure. Therefore, the difference between stationary only and ELM exposure

levels out for the case of high-energy traps, and the content values can even reach the same level at low-flux background exposure (see the case at $q_{\text{stat}} = 1 \text{ MW m}^{-2}$).

The acquired results illustrate the general patterns of the fuel retention in PFCs. However, an increase in the retention with the trap concentration and the barrier for detrapping from them is more important. It suggests that formation of high-energy traps in the material bulk during operation of an ITER-type device (e.g. because of neutron irradiation) can establish conditions for an enhancement in the fuel retention rate under ELM-like exposure. This enhancement can compensate for the reduction in the retention rate induced by an additional heat flux brought by the energetic ELM particles.

3.5. Effect of the D recombination rate

Up to now, the presented results were obtained within the approximation of instantaneous recombination and desorption on the front W surface. Depending on the surface condition, the rate of D release can change, affecting the dynamics of D retention. To illustrate the role of recombination, row (d) in Fig. 6 shows values of the D content at different rates of recombination on the W surface. Additionally, two specific cases are presented for comparison, namely “ ∞ ” showing the data within the instantaneous recombination approximation and $E_c = 0$ corresponding to the assumption of a “clean” surface.

According to the figure, the approximation of instantaneous recombinations fairly reproduces the results obtained with low values of the chemisorption barrier, i.e. almost “clean” surface. At higher values, an increase in the chemisorption barrier leads to a rise in the D content. This increase is observed more apparently in the case of combined exposure since the used recombination coefficient decreases with the temperature. By analysing the data for the highest value of the chemisorption barrier, one can also notice that the results under combined exposure are close to the ones obtained under stationary irradiation only. This observation allows assuming a possible enhancement in the D retention rate under ELM-like conditions in the case of strongly limited desorption from a surface.

4. Analytical expressions for the ELM-average distribution of D at saturation

The previous section presented the numerical results limited to 1000 s of continuous ELM-like exposure. Such simulations are time-consuming because of the time-refinement (small time-step) required to simulate accurately the ELM bursts. In our case (single CPU core of Intel Xeon Gold 6240), the time to simulate 1000 s of exposure with 100 Hz ELMs exceeded 72 h. In a similar study, D. Dasgupta et al. [32] suggested using a stationary ELM-average heat load instead of a time-periodic one to simplify the modelling and reduce the computational cost. However, the authors considered only ELM-like heat loads at a stationary irradiation flux. In this section, we analyse the applicability of an extended approach to simulate the D retention in W within the model, used in the present paper.

4.1. ELM-average heat and particle loads

Transition to ELM-average quantities involves splitting periodic ELM heat and particle fluxes in eqs. (2) and (4a) into slowly varying and fluctuating parts [79]:

$$q_{\text{ELM}}(t) = \bar{q}_{\text{ELM}} + \tilde{q}_{\text{ELM}}(t); \quad (19a)$$

$$\Gamma_{\text{ELM}}(t) = \bar{\Gamma}_{\text{ELM}} + \tilde{\Gamma}_{\text{ELM}}(t), \quad (19b)$$

where $\bar{q}_{\text{ELM}}$ ($\bar{\Gamma}_{\text{ELM}}$) is part of the ELM heat (particle) flux averaged of the ELM period, $\tilde{q}_{\text{ELM}}$ ($\tilde{\Gamma}_{\text{ELM}}$) is the fluctuating part of the ELM heat (particle) flux. ELM-average parts are:

$$\bar{q}_{\text{ELM}} = f_{\text{ELM}} \varepsilon_{\parallel} \sin(\alpha); \quad (20a)$$

$$\bar{\Gamma}_{\text{ELM}} = \frac{\sqrt{\pi}}{2} \Gamma_{0,\text{ELM}} \xi \operatorname{erfc}(\xi), \quad (20b)$$

where $\xi = \tau f_{\text{ELM}}$, $\operatorname{erfc}(x)$ - complementary error function.

Substitution of the decomposed heat flux into the heat equation allows analysing the ELM-average temperature field separately. Even though the material thermal properties depend non-linearly on temperature, the contribution of fluctuating terms to the evolution of the ELM-average material temperature will be small. The induced temperature rise due to a surface heat pulse is proportional to $1/\sqrt{\kappa C \rho}$ [80], which varies weakly with temperature. On the other hand, a complete analytical treatment of the reaction-diffusion system is more difficult considering the non-linearity of rate coefficients.

To simplify the task, one can focus on the case of relatively small ELM bursts, such that the induced fluctuations of temperature and concentration fields are small compared to the ELM-average parts. The latter will mostly determine the temporal evolution of fields up to the quasi-stationary phase. Under the assumption, the solution obtained using ELM-average loads will coincide with the one of a full time-dependent problem, averaged over the ELM period. In the case of large ELMs, terms related to fluctuations will contribute to the evolution of the D content. To distinguish these two cases, one can compare the ELM-average and fluctuating components of temperature and D source terms, included in eqs. (4).

We assume that the radiative heat losses in the heat equation are negligible. With substitution of eq. (19a) into the heat equation and averaging over the ELM period, one can easily obtain an approximate expression for the ELM-average temperature:

$$\bar{T}(x) = T_r + \bar{q}_{\text{tot}} \int_x^L \frac{dx}{\kappa(T)}, \quad (21)$$

where $\bar{q}_{\text{tot}}$ is the total average heat flux at the front surface (see eq. (17b)), T_r is the equilibrium temperature at the rear material side, which is found from the heat flux balance: $\bar{q}_{\text{tot}} = q_{\text{loss}}^*(T_r)$. The peak temperature fluctuation due to a surface heat source can be estimated as:

$$\tilde{T}^{\text{max}} \propto \tilde{q}_{\text{ELM}}^{\text{max}} \sqrt{\frac{\tau}{\kappa C \rho}}. \quad (22)$$

By computing the value of the ELM-average heat flux, one can notice that $\tilde{q}_{\text{ELM}}^{\text{max}} \approx q_{\text{ELM}}^{\text{max}}$. The ELM heat flux reaches the peak value of $q_{\text{ELM}}^{\text{max}} \approx 0.84 q_{0,\text{ELM}}$ at $\tau/t \approx 1.27$. The induced temperature fluctuation can be considered small if $\tilde{T}^{\text{max}} \ll \bar{T}(x=0)$. A representative relation can be obtained assuming that material properties are independent of temperature:

$$q_{0,\text{ELM}} \ll \bar{q}_{\text{tot}} \frac{L}{L_{\text{heat}}} \left(1 + \frac{T_r \kappa}{\bar{q}_{\text{tot}} L} \right), \quad (23)$$

where $L_{\text{heat}} = \sqrt{\kappa \tau / C \rho}$ is the characteristic heat propagation length. For the implantation source terms, a similar relation stands: $\tilde{S}_{\text{ELM}}^{\text{max}} \ll \bar{S}_{\text{tot}}$, where $\bar{S}_{\text{tot}} = S_{\text{stat}} + \bar{S}_{\text{ELM}}$. Similarly to the previous case, the peak flux fluctuation of the particle flux due to an ELM burst is approximately: $\tilde{\Gamma}_{\text{ELM}}^{\text{max}} \approx \Gamma_{\text{ELM}}^{\text{max}}$. The ELM particle flux peaks at $t = \tau$ with the value of $\Gamma_{0,\text{ELM}} / \exp(1)$. By expanding the initial relation and using the defined spatial distribution of sources, one can come to the following condition:

$$\Gamma_{0,\text{ELM}} \ll \Gamma_{\text{stat}} \frac{N_{\text{stat}} \sigma_{\text{ELM}} (1 - r_{\text{stat}})}{N_{\text{ELM}} \sigma_{\text{stat}} (1 - r_{\text{ELM}})}. \quad (24)$$

Both relations simply denote that amplitudes of ELM loads have to be small compared to average loads. Based on the previous section, such conditions can be reached if the ELM frequency is high. Noteworthy, the relations are not independent since heat and particle fluxes are related.

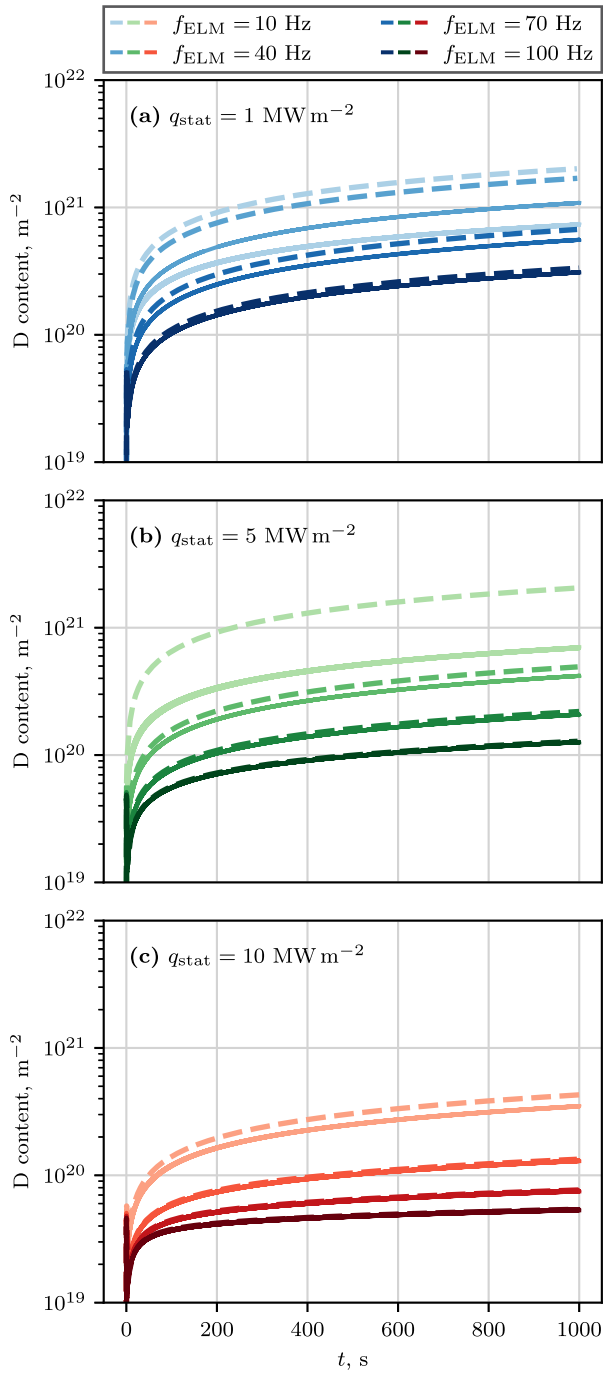


Fig. 7. Comparison between temporal dependencies of the D content obtained with full time-dependent ELM loads (solid lines) and with ELM-average loads (dashed lines). The calculation parameters are $E_{dt} = 1.5$ eV, $\eta_t = 10^{-2}$ at.%, and $E_{ELM} = 2T_{e,ped}$.

To demonstrate the applicability of the approach, an additional set of simulations was performed using ELM-average quantities in eq. (1) and eq. (4a) with homogeneous boundary conditions for the diffusion problem (see eq. (8)). The comparison is shown in Fig. 7, where the results are presented for different exposure conditions. In accordance with the qualitative analysis, the divergence of the results obtained with ELM-average loads from the one of full time-dependent calculations decreases at higher ELM frequencies and higher background exposure loads. In the case of $q_{stat} = 10$ MW m $^{-2}$ (subfigure (c)), the results agree well over the whole range of ELM-frequencies. In addition to the findings of D. Dasgupta et al. [32], the obtained results support the use of ELM-average

quantities for analysing the effect of small ELM-like loads on the fuel retention, demonstrating a simpler analysis method for future studies. Similar to the mentioned work, the computational time in our case was reduced from days to minutes.

4.2. Steady-state approximation

The use of ELM-average loads makes it possible to obtain analytical expressions for the average D distribution at saturation. Following earlier studies [75,81], the derivation is made considering implantation sources S_i in eq. (4a) as point sources at depths X_i . The steady-state approximation and substitution of eq. (5) into eq. (4a) allow recasting eqs. (4) into:

$$\frac{\partial}{\partial x} \left(D(\bar{T}) \left[\frac{\partial \bar{c}_m}{\partial x} + H(\bar{T}) \bar{c}_m \right] \right) = -S_{stat}^{\delta} - \bar{S}_{ELM}^{\delta}; \quad (25a)$$

$$\bar{c}_t = \frac{n_t}{1 + \frac{n_t v_{dt}(\bar{T})}{\bar{c}_m v_t(\bar{T})}}, \quad (25b)$$

where $S_i^{\delta} = \Gamma_i^{imp} \delta(x - X_i)$ with $\delta(x)$ being the Dirac delta function, $H(\bar{T}) = Q(\bar{T}) \partial_x \bar{T} / k \bar{T}^2$. It is important to mention that eq. (25b) can be generalised for the case of several types of trapping sites having a non-uniform spatial distribution. Though, we limit the analysis to the case of one trapping site studied in the work. The solution of eq. (25a) determines the distribution of both species at saturation. It can be acquired assuming the Dirichlet boundary condition (D.C.B.) on the front surface and the Dirichlet/Neumann condition (N.B.C.) on the rear surface. Therefore, the presented further expressions cannot be applied to multi-material geometries, but can be used, for example, to estimate the D content in pure W samples studied in laboratory experiments.

Solving of the stationary diffusion equation involves constructing an appropriate Green function and applying the superposition theorem (see Appendix B). For the used heat of transport ($Q(T) = -0.0045 k T^2$), the H -term is independent of temperature. Thus, the solution of eq. (25a) is:

$$\bar{c}_m = \exp(-Hx) \times \begin{cases} C_1(x, X_{stat}) + C_1(x, X_{ELM}), & x \in [0, X_{stat}] \\ C_2(x, X_{stat}) + C_1(x, X_{ELM}), & x \in [X_{stat}, X_{ELM}] \\ C_2(x, X_{stat}) + C_2(x, X_{ELM}), & x \in [X_{ELM}, L_W] \end{cases} \quad (26a)$$

$$C_1(x, X_i) = \Gamma_i^{imp} G_1(x, X_i); \quad (26b)$$

$$C_2(x, X_i) = \Gamma_i^{imp} G_2(x, X_i), \quad (26c)$$

where the Green function is defined with eqs. (B.6a) for the case of the Dirichlet boundary condition and with eqs. (B.6b) for the case of the Neumann boundary condition.

In Fig. 8, analytical concentration profiles are compared to the ones computed numerically for three different cases:

- Case 1: $q_{stat} = 1$ MW m $^{-2}$, $f_{ELM} = 10$ Hz, $\eta_t = 10^{-3}$ at.%;
- Case 2: $q_{stat} = 5$ MW m $^{-2}$, $f_{ELM} = 50$ Hz, $\eta_t = 10^{-2}$ at.%;
- Case 3: $q_{stat} = 10$ MW m $^{-2}$, $f_{ELM} = 100$ Hz, $\eta_t = 10^{-2}$ at.%,

with $E_{dt} = 1$ eV, $E_{ELM} = 2T_{e,ped}$. It must be noted that the simulated concentration profiles under ELM-average loads were obtained with the Gauss distribution of implantation sources (eqs. (7)). Moreover, the simulations lasted until the saturation was reached. For the sake of illustration, the D retention reaches 99% of steady-state level within ~ 16300 s in Case 2 with the Dirichlet boundary condition imposed on both sides. This time is more than 3 times higher compared to the case with similar exposure conditions without traps [32]. The analytical profiles correlate fairly well with the simulated profiles at different combinations of input parameters, but represent an underestimation, similar to the case

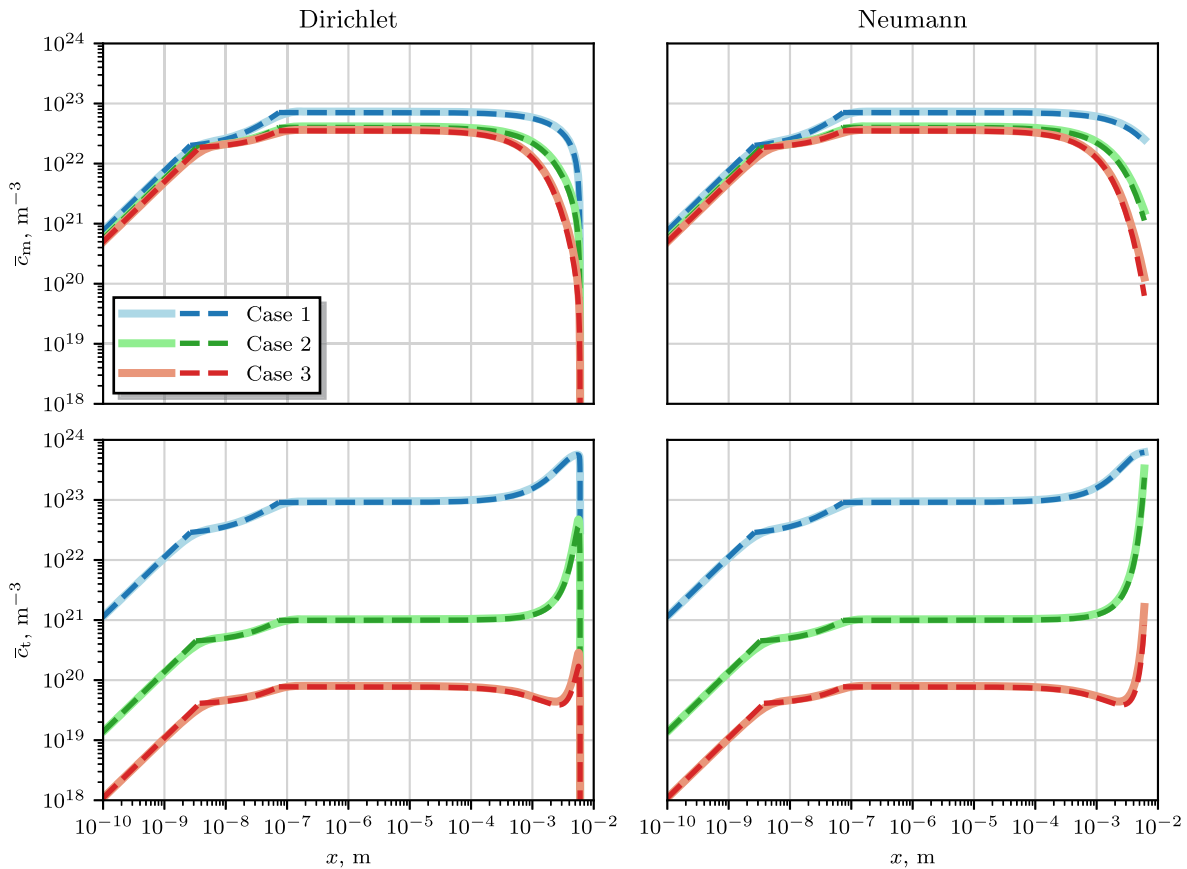


Fig. 8. Comparison of analytical concentration profiles at saturation (dashed lines) with the ones obtained by numerical calculations using ELM-average loads (solid lines).

of homogeneous temperature distribution in [75]. Overall, the analytical model can be used for an estimation of the upper retention limit, bottlenecked by the numerical evaluation of integral expressions.

5. Conclusions

The effect of fast repetitive ELM-like loads on the D retention in W has been analysed over the wide range of exposure, trapping, and desorption conditions. On small time-scales (i.e. after a few ELM-like impulse loads), the onset of transient events is found to increase the D content compared to the case of stationary (background) exposure only. However, longer transient exposure is shown to mainly reduce the D retention rate. This reduction is caused by reaching significantly higher material temperatures due to the arrival of energetic ELM ions. Within the considered conditions, the decrease is more apparent at higher inter-ELM loads and higher frequencies of transients. Therefore, previous estimations of the hydrogen retention in PFCs, which did not account for ELMs, might represent an overestimation in the case of high background loads at the point of an ELM crash (for example, near the strike point of the attached divertor). In the opposite case of low background loads (for instance, the region far away from the strike point or the first wall), neglecting of low-frequency ELMs can result in an underestimation of the retention.

Several factors have been identified capable of enhancing the D retention rate in the presence of ELM-like loads. At the fixed transient heat flux, an increase in the ELM particle flux-to-energy ratio can result in a slightly higher retention rate. The existence/formation of strong trapping sites, limiting the release at higher material temperatures, can also bring the retention up. Moreover, the difference between ELM and stationary only cases is revealed to decline with the transition from weak

to strong traps. The final finding concerns the role of the surface recombination rate. If desorption from a surface is strongly limited, the D accumulation noticeably intensifies under ELM exposure, whereas a more moderate rise is seen in the stationary case. According to the simulations, these effects are more pronounced at lower values of background irradiation fluxes. On top of that, additional heating during transients provides conditions for a faster diffusion into the bulk. The enhanced mobility can affect the propagation time through a PFC, the long-term retention of fuel in the bulk, and permeation flux into the cooling system.

In addition, we have discussed the applicability of using only ELM-average components of loads for the simulations of the D retention dynamics. The approach is shown to be valid when transients induce small perturbations in temperature and concentration fields. Nonetheless, the use of ELM-average quantities drastically reduces the computation cost and opens a way for simpler estimations of the long-term retention under small ELM-like events. The ELM-average components also allowed us to derive approximate expressions for the steady-state D distribution in the presence of a temperature gradient. Analytical expressions agree well with the results of numerical calculations and can be used to estimate the D content at saturation.

CRediT authorship contribution statement

Vladimir Kulagin: Writing – original draft, Visualization, Software, Investigation, Conceptualization. **Yury Gasparyan:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis.

Table A.1
Fitting parameters for eqs. (17).

$q_{\text{stat}}, \text{ MW m}^{-2}$	Parameter	ELM frequency, Hz										
		0 ^a	10	20	30	40	50	60	70	80	90	100
1	a, 1	0.151	0.135	0.134	0.132	0.130	0.129	0.128	0.127	0.126	0.125	0.124
	b, K ⁻¹	0.063	0.012	0.009	0.008	0.007	0.006	0.006	0.005	0.005	0.005	0.005
	c, 1	0.145	0.122	0.126	0.125	0.124	0.123	0.122	0.121	0.120	0.120	0.119
	d, K	0.96	4.16	6.05	7.02	7.82	8.49	9.11	9.65	10.18	10.65	11.09
5	a, 1	0.139	0.131	0.128	0.126	0.125	0.124	0.123	0.123	0.122	0.121	0.121
	b, K ⁻¹	0.013	0.007	0.006	0.005	0.005	0.004	0.004	0.004	0.004	0.004	0.004
	c, 1	0.134	0.120	0.121	0.121	0.120	0.119	0.118	0.117	0.117	0.117	0.116
	d, K	4.29	8.20	9.00	9.83	10.57	11.22	11.81	12.36	12.88	13.34	13.78
10	a, 1	0.130	0.124	0.123	0.121	0.121	0.120	0.119	0.119	0.119	0.118	0.118
	b, K ⁻¹	0.007	0.005	0.004	0.004	0.003	0.003	0.003	0.003	0.003	0.003	0.003
	c, 1	0.125	0.118	0.118	0.117	0.116	0.116	0.115	0.115	0.114	0.114	0.114
	d, K	7.81	11.79	12.31	13.09	13.82	14.48	15.08	15.64	16.17	16.66	17.14

^a Stationary exposure without ELMs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Fitting parameters for the heat flux flowing through the W-Cu interface

For different exposure conditions in Table A.1, fitting of heat fluxes flowing through the W-Cu interface was performed with Levenberg-Marquardt optimisation algorithm.

Appendix B. General solution of the steady-state diffusion equation considering the temperature gradient

To solve the steady-state diffusion equation with a known temperature distribution ($\partial_x T \neq 0$), we perform the same analytical steps as were done for the diffusion problem at uniform temperature [75]. To start with, eq. (25a) is generalised to account for k sources:

$$\begin{cases} \frac{\partial}{\partial x} \left(D(T) \left[\frac{\partial c}{\partial x} + H(T)c \right] \right) = - \sum_{i=1}^k S_i(x); \\ c|_{x=0} = 0; \\ \left[\begin{array}{l} c|_{x=L} = 0, \text{ if D.B.C.}; \\ D(T) \left(\frac{\partial c}{\partial x} + H(T)c \right) \Big|_{x=L} = 0, \text{ if N.B.C.} \end{array} \right] \end{cases} \quad (\text{B.1})$$

Compared to eq. (25a), the notation is reduced for clarity. The diffusion equation can be re-written:

$$\frac{\partial}{\partial x} \left(p(x) \frac{\partial}{\partial x} \left[c e^{\int_0^x H(T) d\chi} \right] \right) = - \sum_{i=1}^k S_i(x), \quad (\text{B.2})$$

with $p(x) = D(T) e^{-\int_0^x H(T) d\chi}$. The solution of this equation can be found by invoking the superposition theorem:

$$c = e^{-\int_0^x H(T) d\chi} \sum_{i=1}^k \left(\int_0^L G(x, y) S_i(y) dy \right), \quad (\text{B.3})$$

where $G(x, y)$ is the Green function satisfying the following auxiliary problem:

$$\begin{cases} \frac{\partial}{\partial x} \left(p(x) \frac{\partial G(x, y)}{\partial x} \right) = -\delta(x - y); \\ G(x, y)|_{x=0} = 0; \\ \left[\begin{array}{l} G(x, y)|_{x=L} = 0, \text{ if D.B.C.}; \\ p(x) \frac{\partial G(x, y)}{\partial x} \Big|_{x=L} = 0, \text{ if N.B.C.}, \end{array} \right] \end{cases} \quad (\text{B.4})$$

with connection conditions:

$$\begin{aligned} G(x, y)|_{x=y^+} &= G(x, y)|_{x=y^-}; \\ p(x) \frac{\partial G(x, y)}{\partial x} \Big|_{x=y^+} - p(x) \frac{\partial G(x, y)}{\partial x} \Big|_{x=y^-} &= -1; \end{aligned} \quad (\text{B.5})$$

For both types of boundary conditions at the rear surface, the Green function can be expressed as:

$$\text{D.B.C. : } \begin{cases} G_1(x, y) = \int_0^x \frac{d\chi}{p(\chi)} \int_y^L \frac{d\chi}{p(\chi)}, & x \in [0, y] \\ G_2(x, y) = \int_0^y \frac{d\chi}{p(\chi)} \int_x^L \frac{d\chi}{p(\chi)}, & x \in [y, L] \end{cases} \quad (\text{B.6a})$$

$$\text{N.B.C. : } \begin{cases} G_1(x, y) = \int_0^x \frac{d\chi}{p(\chi)}, & x \in [0, y] \\ G_2(x, y) = \int_y^L \frac{d\chi}{p(\chi)}, & x \in [y, L] \end{cases} \quad (\text{B.6b})$$

If the temperature gradient is low, i.e. $\partial_x T \rightarrow 0$, $T \rightarrow \text{const.}$, the Green function is reduced to:

$$\text{D.B.C. : } \begin{cases} G_1(x, y) = \frac{x}{D(T)} \frac{L-y}{L}, & x \in [0, y] \\ G_2(x, y) = \frac{y}{D(T)} \frac{L-x}{L}, & x \in [y, L] \end{cases} \quad (\text{B.7a})$$

$$\text{N.B.C. : } \begin{cases} G_1(x, y) = \frac{x}{D(T)}, & x \in [0, y] \\ G_2(x, y) = \frac{y}{D(T)}, & x \in [y, L] \end{cases} \quad (\text{B.7b})$$

which is the exact same representation as obtained in [75]. Substitution of eqs. (B.6) into eq. (B.3) then provides the general solution of the steady-state diffusion problem:

$$c = e^{-\int_0^x H(T)dx} \sum_{i=1}^k \left(\int_0^x G_2(x, y) S_i(y) dy + \int_x^L G_1(x, y) S_i(y) dy \right).$$

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